

SAKON NAKHON AGROMET, Thailand

WMO: 483550

Lat: 17.117N

Lon: 104.050E

Elev: 630

StdP: 14.36

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	52.4	55.8	46.2	47.2	59.5	48.9	52.4	60.5	9.5	76.7	8.0	77.5	0.1	N/A	0.284

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	18.8	100.2	78.4	97.3	78.5	95.1	78.8	82.3	91.6	81.7	90.7	81.1	89.8	2.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.1	159.9	86.5	79.5	156.6	86.1	78.9	153.7	85.8	46.8	90.6	46.1	91.5	45.5	90.6	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.6	6.9	5.7	DB	47.1	102.2	3.1	2.9	44.9	104.3	43.1	106.0	41.4	107.6	39.1	109.7	
			WB	46.0	84.1	3.1	1.0	43.8	84.8	42.0	85.4	40.3	86.0	38.1	86.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.0	71.7	75.9	81.3	84.9	84.3	83.9	82.6	82.2	82.1	80.4	77.5	72.4
	DBStd	6.01	5.20	5.98	5.86	4.97	3.15	2.62	3.07	2.32	2.17	2.54	4.00	5.27
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	23	11	3	1	0	0	0	0	0	0	0	0	8
	CDD50	10934	673	726	971	1048	1064	1018	1010	997	964	943	824	696
	CDD65	5482	219	309	507	598	599	568	545	532	514	478	374	239
	CDH74	56383	1754	2942	5558	7235	6721	6462	5725	5382	5105	4353	3189	1957
Wind	WSAvg	0.9	1.1	1.0	1.1	1.0	0.8	0.9	1.1	1.0	0.6	0.6	0.8	1.0
	PrecAvg	63.6	0.2	0.7	1.9	3.1	9.7	11.8	8.6	13.7	10.4	2.9	0.4	0.2
	PrecMax	79.3	1.9	3.7	5.5	5.0	17.9	17.7	14.3	39.3	19.7	10.6	2.3	1.8
	PrecMin	53.7	0.0	0.0	0.0	0.6	5.4	5.8	2.2	4.2	4.3	0.1	0.0	0.0
	PrecStd	8.3	0.5	1.1	1.6	1.3	3.4	3.7	4.0	8.2	4.6	2.7	0.6	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.0	97.8	102.3	105.5	99.4	96.9	97.5	93.5	93.0	92.9	94.0	92.5
		MCWB	73.1	74.2	76.2	78.5	81.1	80.7	79.3	81.0	81.1	78.6	78.6	75.1
	2%	DB	88.4	94.4	99.1	101.8	96.5	94.2	93.1	91.2	91.2	91.1	91.4	89.7
		MCWB	72.3	74.1	76.2	79.2	80.8	81.2	80.5	80.5	80.3	77.9	77.1	74.0
	5%	DB	86.0	91.7	96.1	98.2	94.1	92.1	90.8	89.5	89.8	89.6	89.3	86.8
		MCWB	71.2	73.9	76.6	79.5	80.7	80.6	80.1	80.2	79.9	77.7	76.0	72.9
	10%	DB	83.1	88.6	93.0	95.1	91.6	90.2	88.5	87.7	88.1	87.9	87.1	83.9
		MCWB	69.8	73.3	76.6	79.0	80.2	80.2	79.7	79.8	79.4	76.7	74.7	71.0
	0.4%	WB	76.7	78.3	80.1	82.8	83.2	82.9	82.5	82.5	81.9	80.7	79.9	76.9
		MCDB	86.3	89.9	91.9	94.1	93.2	91.6	90.9	90.3	89.8	88.7	90.9	88.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	74.7	77.0	79.1	81.6	82.0	81.9	81.3	81.3	81.3	79.6	78.4	75.3
		MCDB	83.7	89.3	91.6	93.8	91.8	90.4	89.2	88.6	88.9	87.5	88.0	87.1
	5%	WB	72.9	75.6	78.2	80.8	81.3	81.3	80.6	80.6	80.6	78.8	77.4	73.7
		MCDB	81.6	87.8	90.8	93.5	90.9	89.4	88.1	87.6	87.9	86.5	86.2	84.5
	10%	WB	71.4	74.2	77.4	79.9	80.7	80.7	79.9	79.9	79.9	78.0	76.1	72.2
		MCDB	80.7	85.8	89.8	92.2	89.9	88.5	86.8	86.3	86.5	85.1	84.1	82.0
Mean Daily Temperature Range	5% DB	MDBR	22.9	23.5	20.8	18.8	14.5	12.1	10.6	10.7	11.6	15.2	19.2	22.4
		MCDBR	24.5	26.4	23.6	21.6	17.8	15.0	14.3	13.6	14.2	17.3	20.9	23.8
		MCWBR	9.9	9.3	7.3	6.9	6.3	5.5	5.7	5.7	5.5	6.3	8.0	9.6
	5% WB	MCDBR	20.9	22.7	21.0	19.1	15.7	13.7	12.5	12.6	13.0	14.9	18.4	22.5
		MCWBR	9.4	9.0	7.9	7.2	6.5	5.8	5.7	5.7	5.6	5.9	7.5	9.5
Clear-Sky Solar Irradiance	taub		0.457	0.569	0.660	0.626	0.510	0.468	0.455	0.466	0.504	0.519	0.456	0.453
	taud		2.143	1.853	1.685	1.802	2.148	2.291	2.324	2.281	2.160	2.085	2.236	2.177
	Ebn at Noon		261	237	220	229	254	262	266	266	256	248	260	258
	Edh at Noon		46	65	79	71	49	42	41	43	49	51	42	44
All-Sky Solar Radiation	RadAvg		1538	1630	1633	1754	1682	1556	1446	1437	1471	1588	1554	1477
	RadStd		92	103	70	86	110	114	133	102	113	111	56	44

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (14 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page